

LANDTECH INNOVATION HACKATHON 2026

PROJECT REPORT

Land Deed Integrity System
(LDIS)

A Web-Based Land Deed Verification and Registration System with Built-in Anti-Corruption Controls

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Abstract

Land registration is one of the most sensitive public services in Bangladesh, because ownership of land directly affects the livelihood, security and legal standing of citizens. Weaknesses in manual verification, dependence on the individual judgement of a single official, and the possibility of unauthorised registration of government-owned (khas) land or land that has already been sold create opportunities for fraud and corruption.

This report presents the **Land Deed Integrity System (LDIS)**, a web-based system that moves the most important integrity checks of land deed registration out of human discretion and into automated, rule-based and database-enforced controls. The system verifies identity, record-of-rights (khatian), previous deeds, heirship (warish) certificates, government-interest records and payment challans; guides applicants through a nine-step application wizard; and gives the Sub-Registrar a structured, jurisdiction-bound review workflow.

The central design goal is that **corruption should not be able to succeed even when an official is willing to cooperate**. For example, if the land belongs to the government, the database refuses every decision except rejection, whether the request comes from a Sub-Registrar or an Administrator. Registration additionally requires a completed 12-point checklist, verified challans, completed biometric status and an atomic, audited finalization transaction. Every action is written to an append-only audit log.

LDIS is built with Next.js 14, React, Tailwind CSS, Supabase (PostgreSQL and Authentication) and Vercel. It is currently a **demonstration prototype** that works on fictional reference data. It is not an official government system, and it does not produce a legally valid registration. This report describes the problem, the proposed solution, the reasons it is a better approach, the system architecture, every major feature, the security design, testing, limitations and future work.

Keywords: land registration, deed verification, anti-corruption, e-governance, Bangladesh, Next.js, Supabase, PostgreSQL, Row Level Security, audit trail.

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Glossary of Terms

The following terms are used throughout the report. Land-related terms are Bangladeshi administrative terms that are transliterated from Bangla.

Table 1: Land administration terms

Term	Meaning
Khatian (Porcha)	The record-of-rights document that lists the recorded owners of a piece of land together with their shares. In this report the terms "khatian" and "porcha" refer to this record.
Dag	The plot number inside a khatian. One khatian may contain several dags, each with its own area and land class.
Shotok	A unit of land area equal to one hundredth of an acre. All areas in the system are calculated in shotok.
Deed (Dolil)	The legal document that records a transfer of land, for example a sale or a gift.
Baya deed	The previous (parent) deed through which the current owner obtained the land. The chain of baya deeds shows how the land passed from owner to owner.
Donor / Donee	The donor is the person transferring the land (seller). The donee is the person receiving it (buyer).
Warish certificate	A certificate of heirship that lists the legal heirs of a deceased person and the share of each heir.
Khas land	Land owned by the government. Private registration of such land is prohibited.
Challan	A payment slip proving that a fee (for example stamp duty, registration fee or source tax) has been paid to the government.
Attestation	A formal statement that two differently written names (for example due to spelling variation) refer to the same person.
Affidavit	A sworn written statement. In LDIS it is the strongest identity evidence and is used when names differ widely.
Sub-Registrar	The government official who reviews and registers deeds in a sub-district (upazila) office.
Upazila / District	Administrative units of Bangladesh. A Sub-Registrar office is identified by its district and upazila.
NID	National Identity Card, the national identification record of a citizen.
Biometric	Fingerprint or similar identity capture performed at the registration office on the appointment date.

Table 2: Technical terms

Term	Meaning
Web application	A program that runs in a web browser and does not need to be installed on the user's computer.
Next.js / React	JavaScript technologies used to build the pages and server logic of the web application.
Tailwind CSS	A styling framework used to design the visual appearance of the interface.
Supabase	A backend platform that provides a PostgreSQL database, user authentication and storage.
PostgreSQL	The relational database in which all records and business rules are stored.
Vercel	The cloud platform on which the application is deployed and made publicly available.

Term	Meaning
Authentication	The process of proving who a user is, normally by email and password.
Authorization	The process of deciding what an authenticated user is allowed to do.
Role-based access control (RBAC)	A method of allowing actions according to the role of a user (User, Sub-Registrar, Administrator, Data Entry Operator).
Row Level Security (RLS)	A PostgreSQL feature that restricts which rows of a table a user can read or change. It is enabled on LDIS tables.
RPC (Remote Procedure Call)	A database function that the application calls to perform a controlled operation. LDIS uses RPCs so that rules run inside the database.
SECURITY DEFINER	A PostgreSQL option that makes a function run with the permissions of its owner, so that it can access protected data in a controlled manner.
Migration	A versioned SQL file that creates or changes database structure and rules in a repeatable order.
Middleware	Code that runs before a page loads, used here to maintain the login session.
API route / Route handler	A server endpoint that receives a request from the browser and returns data.
Atomic transaction	A group of database operations that either all succeed or all fail together, so partial results are never saved.
Row lock	A database lock that prevents two users from changing the same record at the same time.
Audit log	A permanent record of who did what and when. In LDIS the log is append-only: entries cannot be edited or deleted.
Levenshtein distance	A measure of how different two text strings are, based on the number of single-character edits needed to turn one into the other. It is used to compare names.
Jurisdiction	The area (district and upazila) for which a Sub-Registrar is authorised to act.
Prototype / Demo	A working model built to demonstrate an idea. It uses fictional data and is not an official service.

1. Introduction

1.1 Background

Land is the most valuable asset for most families in Bangladesh, and land disputes are among the most common causes of civil litigation and social conflict. Every land transfer must pass through a registration process in which several facts must be established: who the real owner is, whether the seller is actually entitled to sell the quantity of land offered, whether the same land has already been sold, whether the land is government property, and whether all fees have been paid.

In a conventional process, these facts are checked manually by staff using paper records and personal judgement. When the checking depends on the discretion of one person, the quality and honesty of the result also depend on that person. This creates opportunities for error, delay, and corruption.

1.2 Problem Statement

There is no integrated system that automatically cross-checks identity, ownership, previous sales, heirship, government-interest and payment evidence at the moment of registration, and that prevents an unlawful registration even when the responsible official is willing to approve it. The problem is therefore not only technical efficiency but the structural dependence of an important legal act on the integrity of individuals.

1.3 Objectives

- To design a system in which critical integrity checks are enforced by rules and not by personal discretion.
- To prevent registration of government-owned (khas) land regardless of the decision-maker.
- To detect name mismatches, oversold land, duplicate or conflicting deeds, invalid heirship and invalid challans before registration.
- To give applicants a guided, transparent application process with status tracking.
- To give Sub-Registrars a structured review workflow limited to their own jurisdiction.
- To keep a permanent, tamper-resistant record of every decision for accountability.

1.4 Scope

LDIS covers three connected parts: (1) a verification module for reference lookups, (2) an application module in which a user prepares and submits a deed application, and (3) a staff module in which a Sub-Registrar reviews and finalizes the application and an Administrator supervises the system. The system is a prototype that uses fictional reference data; integration with official government databases, payment gateways and biometric devices is outside the present scope and is described as future work.

1.5 Organization of the Report

Chapter 2 analyses the problem. Chapter 3 presents the proposed solution and Chapter 4 explains why it is better than the manual approach. Chapters 5 and 6 describe the architecture and the database. Chapter 7 describes every feature of the system. Chapter 8 explains the anti-corruption and security controls in detail. Chapters 9 to 12 cover testing, deployment, limitations and future scope, and Chapter 13 concludes the report.

2. Problem Analysis

2.1 The Land Registration Process in Brief

A land transfer normally involves an applicant (donor and donee), a deed, supporting documents (NID, khatian/porcha, previous deed, heirship certificate where land is inherited), payment of fees, a visit to the Sub-Registrar's office, identity capture, and finally the registration of the deed by the Sub-Registrar. The Sub-Registrar is the key decision-maker for the whole sequence.

2.2 Identified Problems

The following problems are commonly associated with manual land registration. They are the problems that LDIS is designed to address.

Table 3: Problems in manual land registration

No.	Problem	Description
P1	Registration of government (khas) land	Land that belongs to the government can be registered privately if the official who reviews the file overlooks, ignores or is induced to ignore the government record.
P2	Dependence on one official	The final decision rests on one Sub-Registrar. An unethical official can approve a defective file and there is no automatic barrier.
P3	Identity and name mismatch	The name of the owner in the khatian may differ from the NID record by spelling, order or deliberate alteration. Manual comparison is subjective.
P4	Double selling and overselling	The same plot, or more area than the plot or the owner's share, may be sold again because earlier sales are not systematically checked.
P5	Heirship and share errors	Inherited land requires correct heir shares. Errors or forged heirship certificates are difficult to detect by sight.

No.	Problem	Description
P6	Fake or reused payment challans	A forged challan, a wrong amount, or a challan already used for another deed can be presented as proof of payment.
P7	Weak accountability	Paper decisions can be changed or lost, and it is hard to prove later who approved what and on what basis.
P8	Lack of transparency for the applicant	Applicants often do not know which documents are required, what fees are due, or why their application is delayed.
P9	Jurisdiction misuse	An official might process applications that do not belong to his or her office.

2.3 Existing Digital Efforts and the Gap

Digital services such as online viewing of land records and online mutation services address parts of land administration. To the best knowledge of the authors, however, there is no integrated pre-registration verification system in operation that (a) cross-checks all the evidence listed above at the point of registration, and (b) technically prevents registration when a critical rule is violated, even if the decision-maker wishes to approve. LDIS has been designed to fill this gap. This statement reflects the authors' knowledge at the time of writing and should be confirmed against the latest government publications before submission.

3. Proposed Solution

3.1 Overview of LDIS

LDIS is a single web platform in which the entire pre-registration and registration workflow takes place. A citizen creates an account, verifies land and identity information against reference records, completes a nine-step application, pays fees against verified challans, books an appointment, and tracks the application. The Sub-Registrar reviews the application inside the same system using a structured checklist. The Administrator supervises users, staff and audit records.

3.2 Design Principles

- **Rules over discretion:** integrity checks are coded as rules; a person can add judgement but cannot remove a rule.
- **Enforcement in the database:** the most critical rules run inside PostgreSQL functions and triggers, so that bypassing the web interface does not bypass the rule.
- **Evidence before approval:** approval is impossible until all required evidence (checklist, challans, biometric status, attestation or affidavit where necessary) is recorded.
- **Explainable warnings:** every risk indicator states what was found, the data source, and that it is a warning and not a legal verdict.
- **Least privilege:** the browser never reads the reference data directly; each role sees only what it needs.
- **Permanent accountability:** every sensitive action is logged in an append-only audit log.

3.3 How Each Problem Is Solved

Table 4: Problem-to-solution mapping

No.	Problem	LDIS Solution
P1	Khas land registration	The system matches location, khatian and dag against the government-interest (khas) table. If a match exists, every decision except Reject is refused by the database for Sub-Registrar and Admin alike, and the rejection must record a reason.

No.	Problem	LDIS Solution
P2	Dependence on one official	Approval requires the 12-point checklist, no system blockers, completed biometric status and verified challans. Rules run in the database and cannot be overridden through the interface.
P3	Name mismatch	Bangla-aware name normalization and similarity scoring with a provisional 90% threshold. Scores between 50% and 90% require a verified attestation; scores below 50% require the original affidavit.
P4	Double selling	Dag history shows all earlier deeds; the system computes entitled and remaining land and raises High-risk warnings when sales exceed the dag area or the donor's entitlement.
P5	Heirship errors	Warish certificates are looked up and heir shares are used in the remaining-land calculation; incorrect share totals are flagged.
P6	Fake or reused challans	Each challan is matched on number, amount and fee type against the reference challan records, and a challan already used by another application is rejected.
P7	Weak accountability	Append-only audit logs and status history record the actor, role, action, note and time of every decision.
P8	Lack of transparency	A guided wizard, status explanations in Bangla and English, correction messages, notifications and a visible appointment.
P9	Jurisdiction misuse	A Sub-Registrar can see and decide only applications whose office district and upazila match the officer's own profile.

3.4 End-to-End User Journey

The journey that every application follows is:

- **Step 1 – Sign up and log in.** New accounts are always ordinary users and require administrator approval.
- **Step 2 – Verify.** The user may use the verification tools to check land, person, deed, heirship and challan information.
- **Step 3 – Apply.** The user completes the nine-step wizard (donor and donee, land, ownership basis, dags and area, class and price, fees and challans, documents, appointment, review and submit).
- **Step 4 – Automated checks.** The system recomputes derived values on the server and attaches blockers and risk flags.
- **Step 5 – Review.** The Sub-Registrar of the matching upazila reviews the application with the 12-point checklist.
- **Step 6 – Decision.** The officer requests correction, rejects, or approves; government-interest cases can only be rejected.
- **Step 7 – Biometric and evidence.** Appointment and biometric status, and attestation or affidavit where required, are recorded.
- **Step 8 – Finalization.** An atomic database transaction creates the registration record with a unique reference and an audit entry.

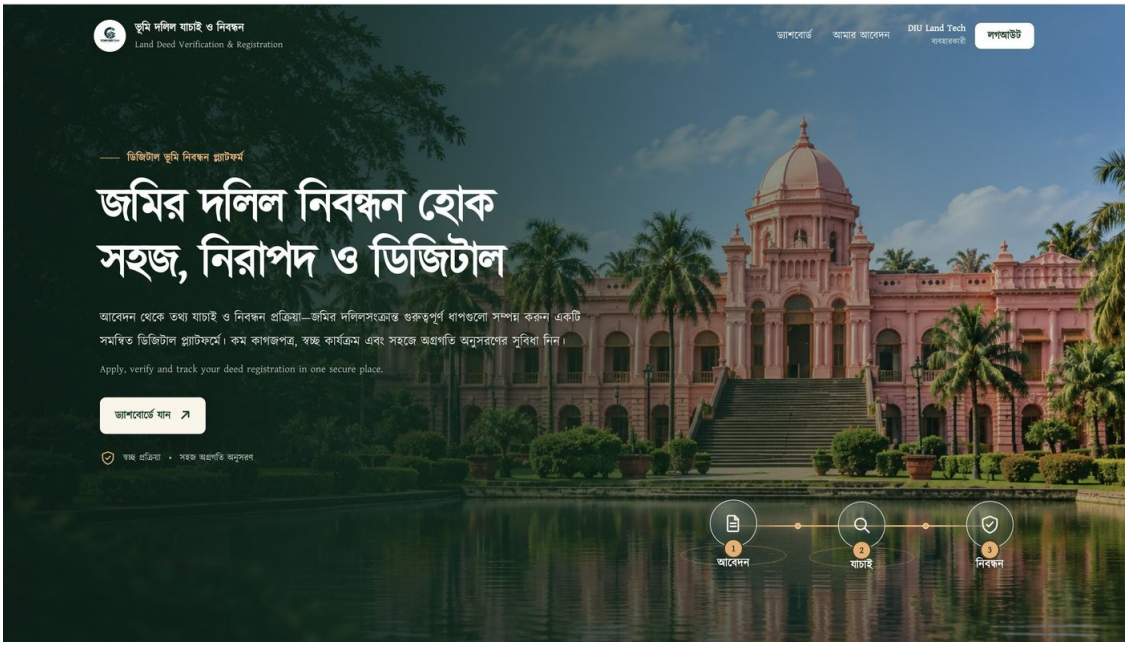


Figure 1: Home page of LDIS

4. Why This Solution Is Better

4.1 Comparison with the Manual Process

Table 5: Manual process compared with LDIS

Aspect	Conventional manual process	LDIS
Government land check	Depends on the official noticing the record	Automatic match against the khas table; only Reject is permitted
Final decision	One person's judgement	Judgement plus mandatory rules enforced by the database
Name comparison	Visual and subjective	Normalized, scored and compared against a fixed threshold
Previous sales	Hard to trace across deeds	Dag history and remaining-land calculation
Payment proof	Paper challan can be forged or reused	Matched with reference record; reuse is blocked
Accountability	Paper trail that can be altered	Append-only audit log and status history
Jurisdiction	Relies on office practice	Enforced by district and upazila match in the database
Applicant experience	Unclear steps and status	Guided wizard, tracking and notifications
Bypass resistance	A willing official can bypass checks	A willing official cannot bypass database rules through the application

4.2 Why Enforcement in the Database Matters

Many systems apply rules only in the user interface, which can be bypassed by calling the server directly. In LDIS the rules are written as PostgreSQL functions and triggers. The web application calls these functions, and the functions themselves check the caller's role, jurisdiction, application state and evidence. As a result, a hidden button, a modified request or a forged role in the browser does not change the outcome.

4.3 Why the Design Resists Corruption

Corruption normally succeeds in two ways: by persuading an official to ignore a rule, or by removing the evidence afterwards. LDIS addresses both. The first is addressed by making the key rules non-negotiable: a government-interest match, an incomplete checklist, an unverified challan, a missing biometric record or a missing attestation each stop the approval no matter who requests it. The second is addressed by the append-only audit log and registration register, in which normal database operations cannot update or delete rows.

This is a preventive approach, not only a detective one. The system does not merely record that an unlawful registration happened; it stops the registration from happening.

4.4 Additional Advantages

- **Speed and convenience:** the applicant can prepare most of the work online and see the status at any time.
- **Consistency:** the same rule is applied to every application.
- **Explainability:** each warning shows its reason and source, which helps the reviewing officer and reduces disputes.
- **Honest limits:** missing data is shown as "unknown" and never as "clear". For example, "no government record found" is displayed with the warning that this is not clearance.
- **Low cost:** the system uses widely available web technologies and a managed backend.

5. System Architecture

5.1 User Roles

Table 6: Roles in LDIS

Role	Home page	Responsibilities
User	/dashboard	Verify records, create and submit applications, upload documents, book appointments, track status and respond to correction requests.
Sub-Registrar	/sub-registrar	Review applications of own district and upazila, complete the 12-point checklist, request corrections, reject, approve, record biometric status, verify attestation and affidavit, and finalize registration.
Administrator	/admin	Approve user accounts, view statistics, the queue of all applications, registrar list and the audit log.
Data Entry Operator	/data-entry	Enter reference data manually into the reference tables through a controlled form.

The role is stored in the profiles table of the database. It is never taken from a form field or from user metadata. Public signup can create only the ordinary User role; the Sub-Registrar and Administrator roles are provisioned by an authorised database administrator.

5.2 Technology Stack

Table 7: Technology stack and the reason for each choice

Layer	Technology	Reason
Frontend and server	Next.js 14 (App Router), React 18, JavaScript	Server-side rendering, route handlers and a single code base for pages and server logic.
Styling	Tailwind CSS	Fast, consistent and responsive interface design.

Layer	Technology	Reason
Database	PostgreSQL (Supabase)	Reliable relational database with RLS, triggers, functions and transactions.
Authentication	Supabase Auth with @supabase/ssr	Secure session handling for server and browser.
Hosting	Vercel	Simple deployment from GitHub with HTTPS.
Version control	GitHub	Source history and collaboration.
Testing	Node.js test runner, SQL check scripts	Unit tests for logic and manual SQL checks for security rules.

5.3 Layered Architecture

The browser communicates only with the Next.js application. Next.js middleware refreshes the user session, and route handlers call database functions using the logged-in user's session. The database functions repeat the role and jurisdiction checks. The reference schema is reached only through limited verification functions.

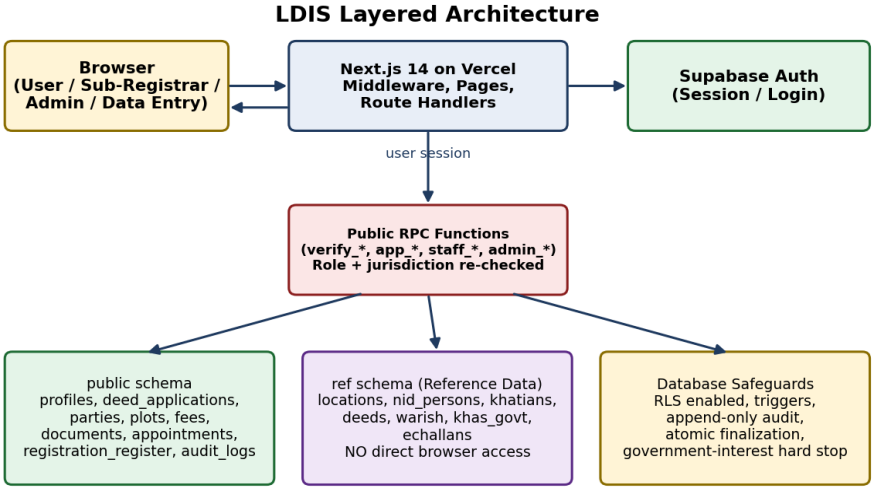


Figure 2: Layered architecture of LDIS

5.4 Request Flow Example

When a Sub-Registrar presses the Approve button: (1) the browser sends the request to the staff route handler, (2) the route handler verifies that the user is logged in and has an approved staff role, (3) the handler calls the database function staff_decide_application, (4) the function locks the application row, confirms the officer's jurisdiction, checks the government-interest rule, the checklist, blockers, biometric status, name-match evidence and challans, and (5) if every condition passes it records the decision, a status history row, an audit entry and a user notification. If any condition fails the function raises an error and nothing is saved.

6. Database Design

6.1 Two Schemas

The database is divided into two schemas. The **ref** schema holds imported reference data that represents the official records the system verifies against. The **public** schema holds the data created by the system itself: user profiles, applications and decisions. The ref schema is protected: RLS is enabled, no public policies exist and no table is granted to browser roles. It can be reached only through limited SECURITY DEFINER verification functions.

6.2 Main Tables

The ref schema contains locations, NID persons, khatians with owners and dags, deeds, warish certificates and heirs, name attestations, government-interest (khas) records and reference challans. The public schema contains profiles (role, approval and jurisdiction), deed applications, parties, ownership, plots, fee items, challans, documents, appointments, status history, corrections, identity evidence, the registrar review checklist, the registration register, audit logs and notifications.

6.3 Key Database Functions

- **Verification functions (verify_...):** controlled lookups into the ref schema.
- **Application functions (app_...):** create, update, submit, withdraw and recompute applications with ownership and state checks.
- **staff_queue, staff_application_detail:** jurisdiction-filtered views for officers.
- **staff_decide_application:** all staff decisions, with government-interest, checklist and evidence rules.
- **staff_save_review_checklist:** saves the 12 checklist decisions and writes an audit entry.
- **staff_finalize_registration:** atomic registration with row lock and preconditions.
- **admin_summary:** administrator-only statistics and recent audit data.

7. Features of the System

This chapter describes the facilities available in each part of the system.

7.1 Authentication and Account Approval

- **Signup and login pages** (/signup, /login) with validation and safe redirect after login.
- **Ordinary user role only:** a signup form cannot create a Sub-Registrar or Administrator.
- **Administrator approval:** a new user stays on a pending-approval page until an Administrator approves the account.
- **Access-denied page:** a user who opens a page of another role is stopped and shown a clear message.
- **Session middleware:** the login session is refreshed on every request.

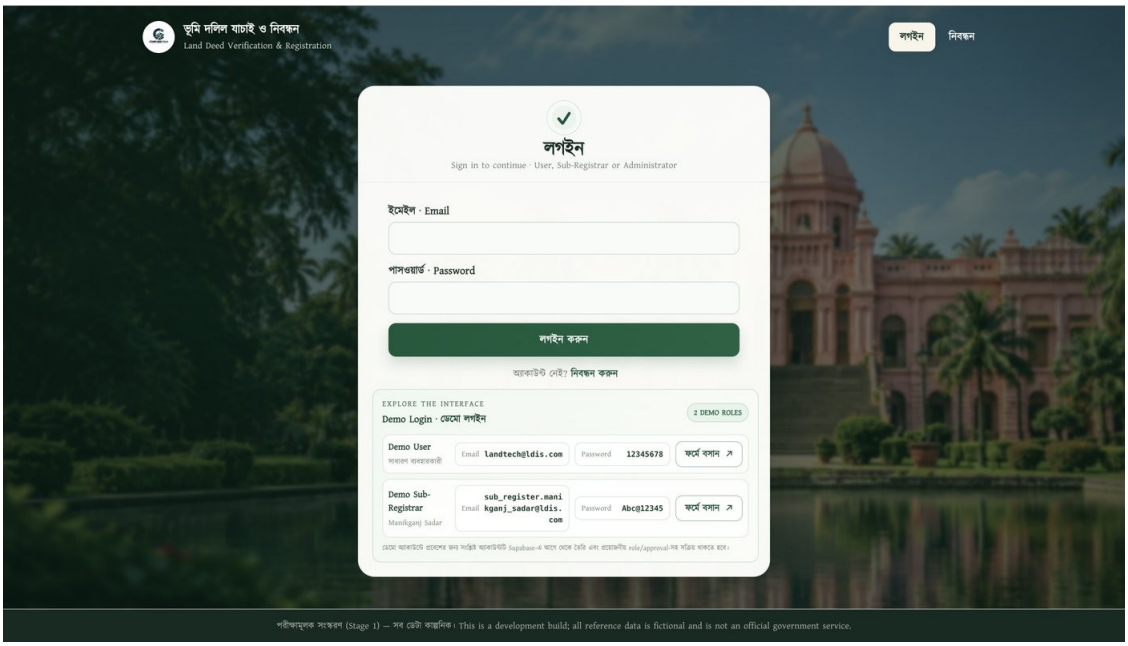


Figure 3: Login page with demo accounts

7.2 User Dashboard

The dashboard is the home page of an ordinary user. It shows the profile, shortcuts to the verification tools, and the list of applications with their current status. A new application can be started from here, and a draft can be continued at any time.

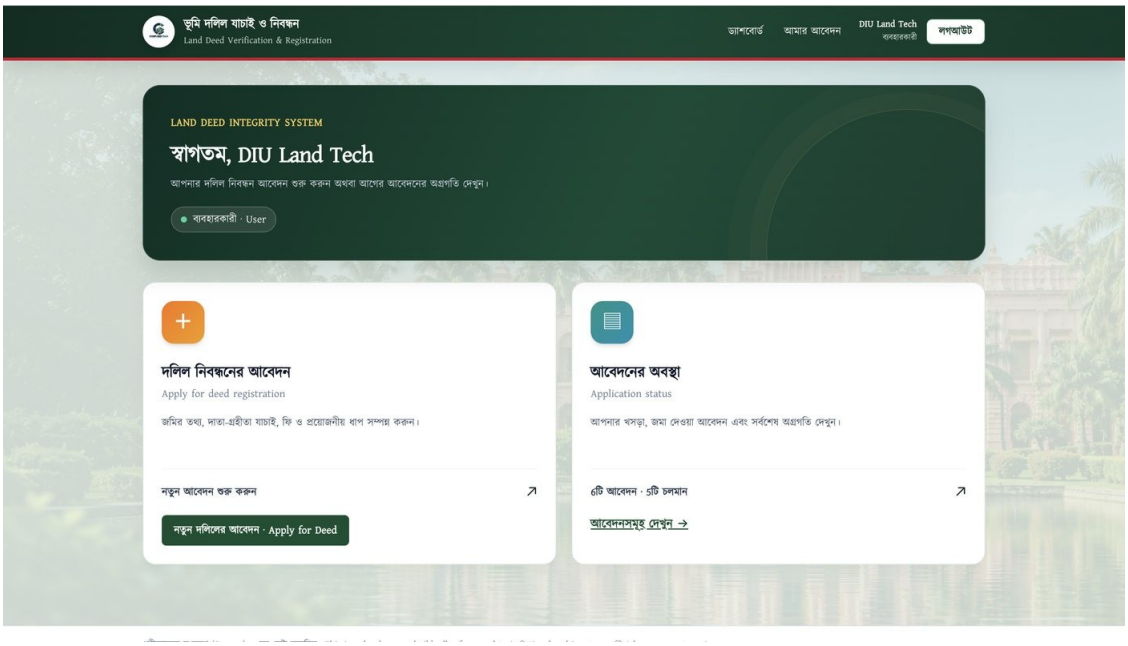


Figure 4: User dashboard

7.3 Verification Tools

The verification module (/verify) lets a user or officer check information against the reference data before or during an application. All lookups go through server routes and controlled database functions. Results are shown in Bangla and English with risk indicators.

Table 11: Verification tools

Tool	Page	What it does
Land verification	/verify/land	Select location (district, upazila, mouza), khatian type and number or dag number. Shows owners and shares, dags, dag history of earlier deeds, owner identity comparison and government-interest result.
Person verification	/verify/person	Looks up an NID record and its identity fields.
Deed verification	/verify/deed	Checks a deed by number and date; reports not found, multiple matches or the same number on other dates.
Warish verification	/verify/warish	Looks up a heirship certificate and checks that the heir shares total correctly.
Challan verification	/verify/challan	Checks a challan by number, amount and fee type against the reference records and shows whether it is already used.

Special facilities: a cascading location picker; Bangla and English digit normalization so that numbers typed in either script match; lookup by dag number to find the khatian; and explanation text that states clearly when a result is only a warning.

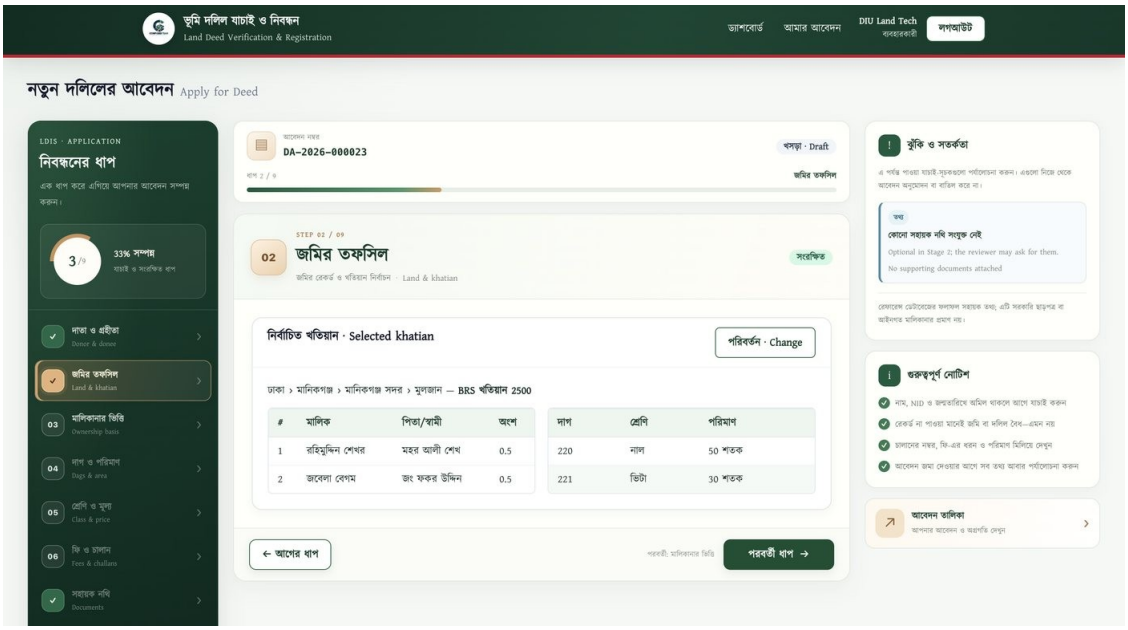


Figure 5: Land and khatian selection with owners and dags (Wizard Step 2)

7.4 Risk Indicator System

Verification results are converted into explainable indicators. Each indicator has a level, a bilingual message, a detail, and a source. Indicators are for display and decision support only: they never approve or clear anything automatically.

Table 12: Indicator levels and examples

Level	Meaning	Examples
High	Hold the case for review	NID and porcha mismatch below 90%; dag oversold; donor already sold more than entitlement; proposed sale exceeds remaining land; government-interest record found; several deeds with the same number and date.
Medium	Needs attention	Owner shares do not total 1; other donors have deeds on the same dag.
Info	Context only	Provisional identity match; donor has earlier sales; same deed number exists on other dates; attestation exists.
Unknown	Data gap, not clearance	No government record found in the supplied database; deed or warish not found; identity data missing.

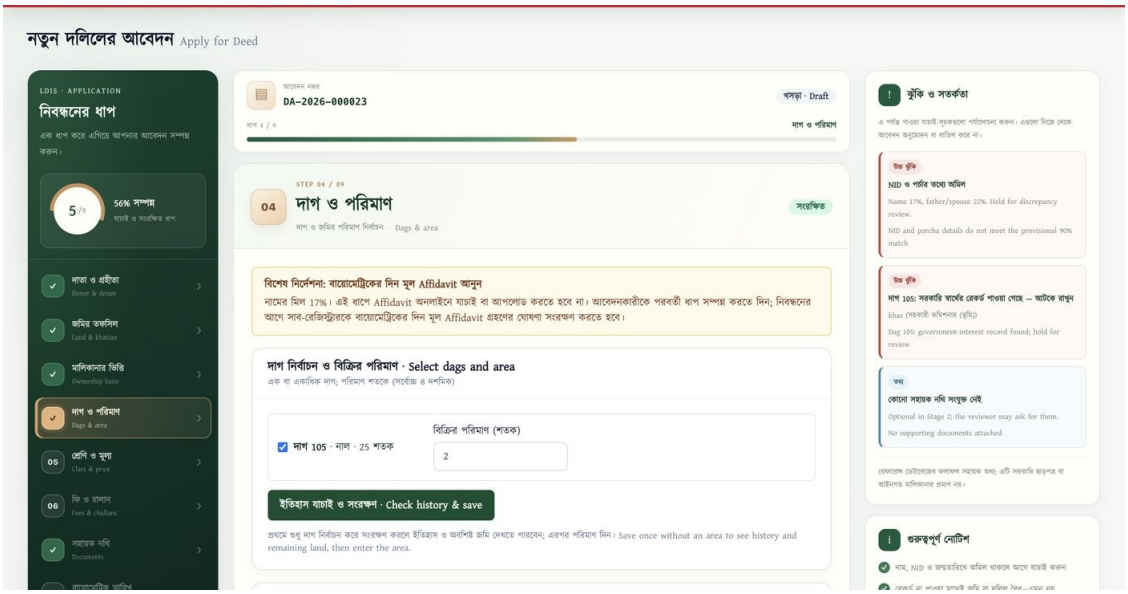


Figure 6: High-risk indicators: NID and porcha mismatch, and a government-interest record (Wizard Step 4)

7.5 Name Matching Algorithm

To compare the owner's name in the porcha with the NID name, the system normalizes the text (Unicode normalization, removal of zero-width joiners and extra spaces, and conversion of Bangla digits) and computes a similarity score based on the Levenshtein distance. The name and the father or spouse name are scored separately.

Table 13: Name match rules

Score	Outcome
90% or higher	Provisional match. This does not prove legal ownership; it only means no discrepancy review is triggered.
50% to below 90%	Discrepancy. The applicant must upload an attestation, and the Sub-Registrar must verify it before approval.
Below 50%	Serious discrepancy. The Sub-Registrar must record receipt of the original affidavit on the biometric day before approval.
Missing data	Marked "insufficient data" and sent for review; never treated as a match.

The 90% threshold is provisional and must be confirmed by the competent authority before any real use.

7.6 Remaining Land Calculation

For every dag, the system computes the land that the donor is entitled to and the land that is still available. The entitled area equals the dag area multiplied by the owner's recorded share (and by the heir's share when ownership comes through a warish certificate). Earlier sales by the donor on the same dag are subtracted. If the proposed sale is larger than the remaining area, a High-risk indicator is raised and the application is blocked from proceeding. The assumptions of this calculation (for example that the same share applies on every dag, and that partitions and cancelled deeds are not modelled) are displayed together with the result.

7.7 Nine-Step Application Wizard

The application wizard (/dashboard/applications) guides the user through nine steps. Progress is saved as a draft, and every derived value (fees, areas, flags, blockers) is recomputed on the server so that a user cannot enter a false calculated value.

Table 14: Wizard steps

Step	Name	Content and facilities
1	Donor and donee	Identity of the donor and one or more donees by NID; identity comparison against NID records.
2	Land and khatian	Selection of location, khatian type and khatian; determines the registration office by district and upazila.
3	Ownership basis	Ownership as the recorded owner or through heirship (warish), name-match result, and attestation requirement.
4	Dags and area	Selection of dags and sale area with remaining-land check.
5	Class and price	Land class (for example নাল, বাড়ি, ভিটা, মাটিয়াল, পুকুর) and sale price.
6	Fees and challans	Automatically calculated stamp fee, registration fee, source tax and other fees; entry and verification of challans; automatic challan numbering.
7	Documents	Upload of NID copy, porcha copy, warish certificate, previous deed, attestation and other documents to private storage.
8	Appointment	Booking of a biometric appointment from available slots and capacity; holidays are respected.
9	Review and submit	Summary of all data and blockers; final submission.

Each step also shows blockers that are returned by the database, so the user knows exactly what must be corrected before the application can be submitted. A deed-nature gate asks for the nature of the deed first, and a witness and numbering rule is applied to the deed.

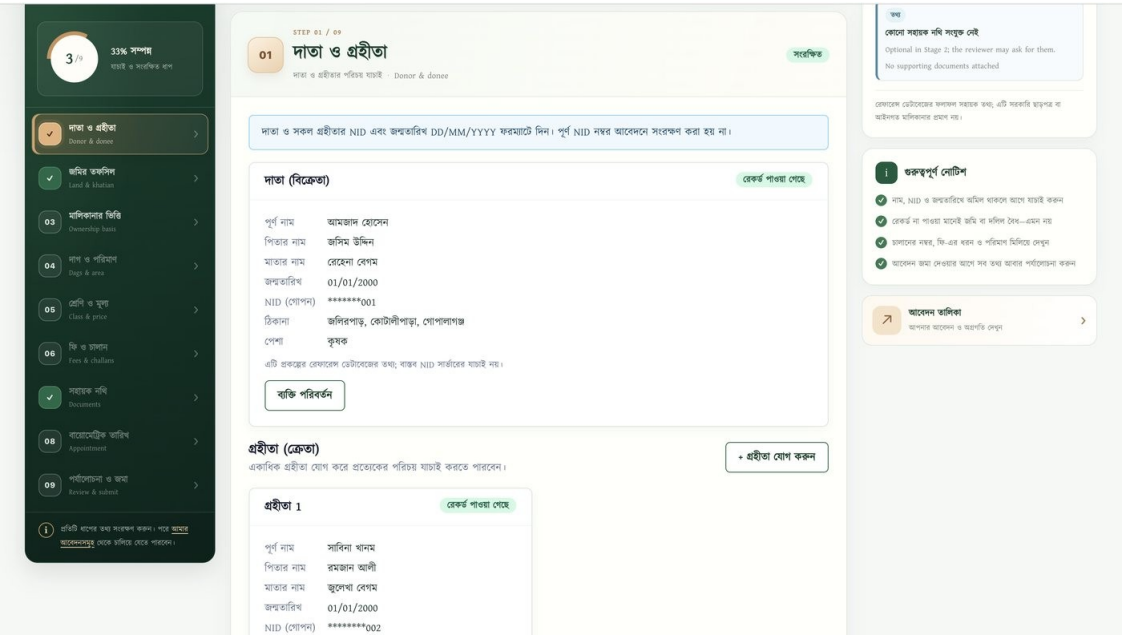


Figure 7: Wizard Step 1: Donor and donee identity check

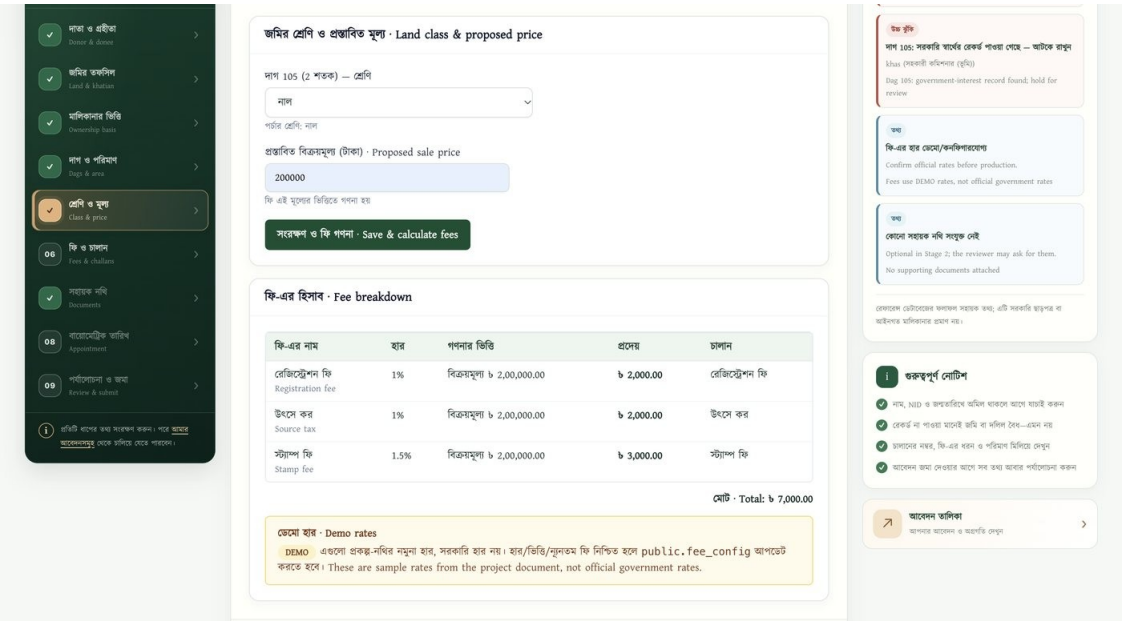


Figure 8: Wizard Steps 5 and 6: Land class, price and fee breakdown

7.8 Application Tracking and Status Lifecycle

After submission, the user can follow the application at /dashboard/applications/[id]. Each status has a Bangla and English explanation so that the applicant understands the situation.

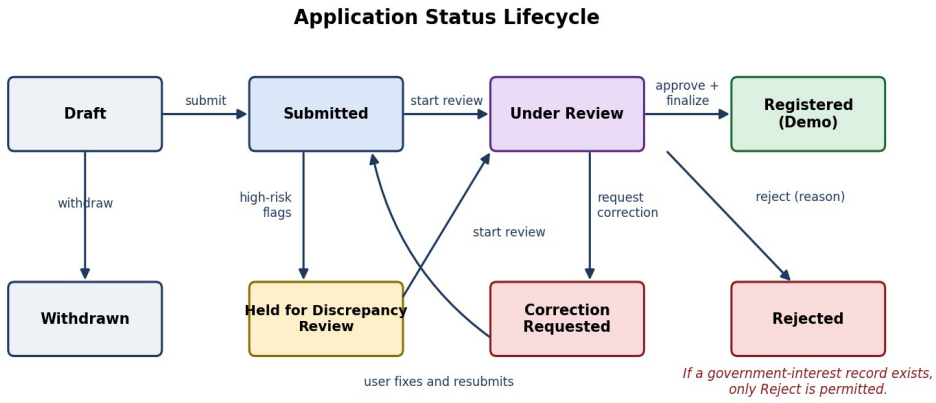


Figure 9: Application status lifecycle

Other facilities of the tracking page include the appointment panel (change or view), the fee table, the list of uploaded documents, the history of status changes, correction messages and in-system notifications.

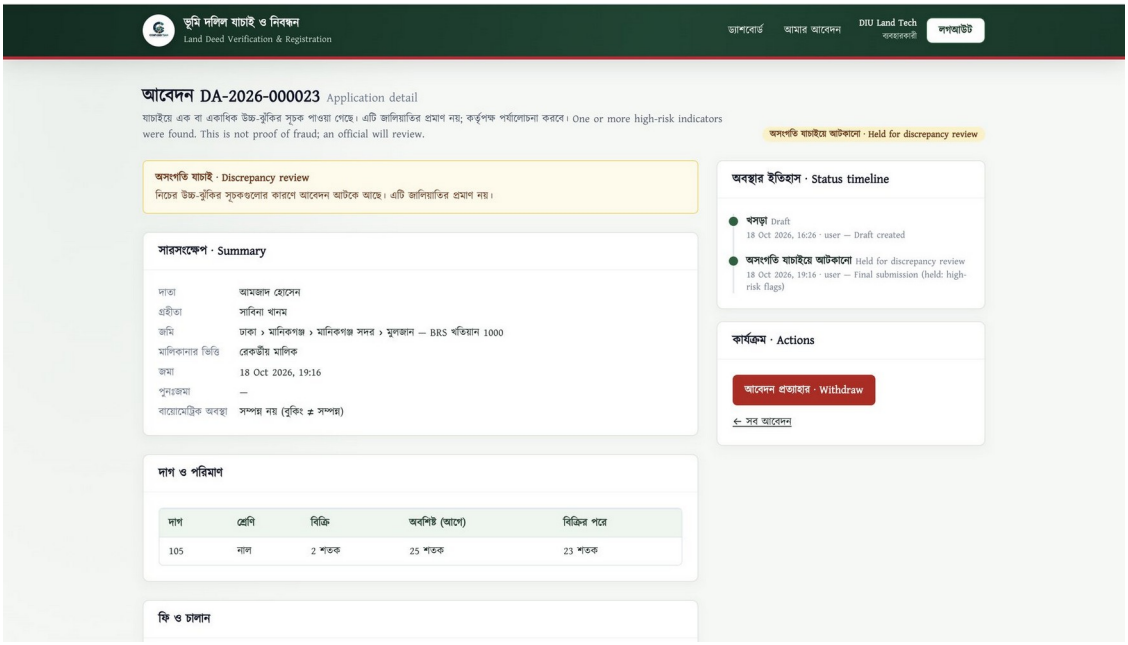


Figure 10: Application detail and status

7.9 Sub-Registrar Panel

The Sub-Registrar panel (/sub-registrar) is available only to approved Sub-Registrars. It shows the list of deeds (applications) of the officer's own upazila. The detail page (/sub-registrar/applications/[id]) shows the full application, the risk flags and the evidence.

- **Jurisdiction-filtered deed list:** only applications whose office district and upazila match the officer's profile are visible.
- **Start review:** moves the application to Under review.
- **12-point review checklist:** every item must be saved as Correct or Incorrect; an Incorrect item requires a written reason.
- **Request correction:** requires a written explanation and optionally the steps that need correction.
- **Confirm appointment and record biometric result:** completed or failed.
- **Verify attestation / record affidavit:** required for name mismatches, performed only by the Sub-Registrar.
- **Approve and finalize:** possible only when all preconditions are met.
- **Reject:** always possible, and mandatory for government-interest cases.
- **Printable registered deed** after finalization.

Table 16: The 12-point Sub-Registrar checklist

No.	Checklist item
1	Application number, applicant details and deed nature are correct.
2	Donor and donee names and identity details are correct.
3	Donor and donee NID and date of birth are correct.
4	Khatian and basis of ownership are verified.
5	The chain of baya (previous) deeds is consistent.
6	Dag numbers, land area and remaining land are correct.
7	Land class and location are correct.
8	Government-interest and restriction information has been reviewed.
9	Required supporting documents have been reviewed.

No.	Checklist item
10	Fee calculation is correct.
11	Challan number, amount and payment verification are correct.
12	Appointment, biometric and registration-related information is correct.

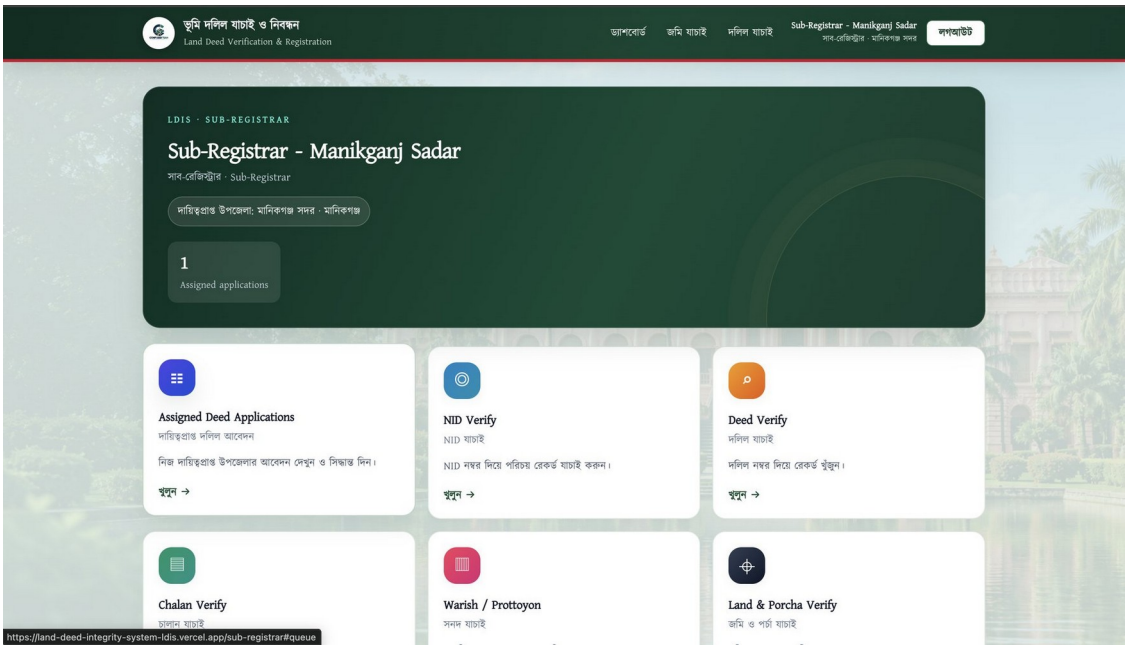


Figure 11: Sub-Registrar dashboard (deed list)

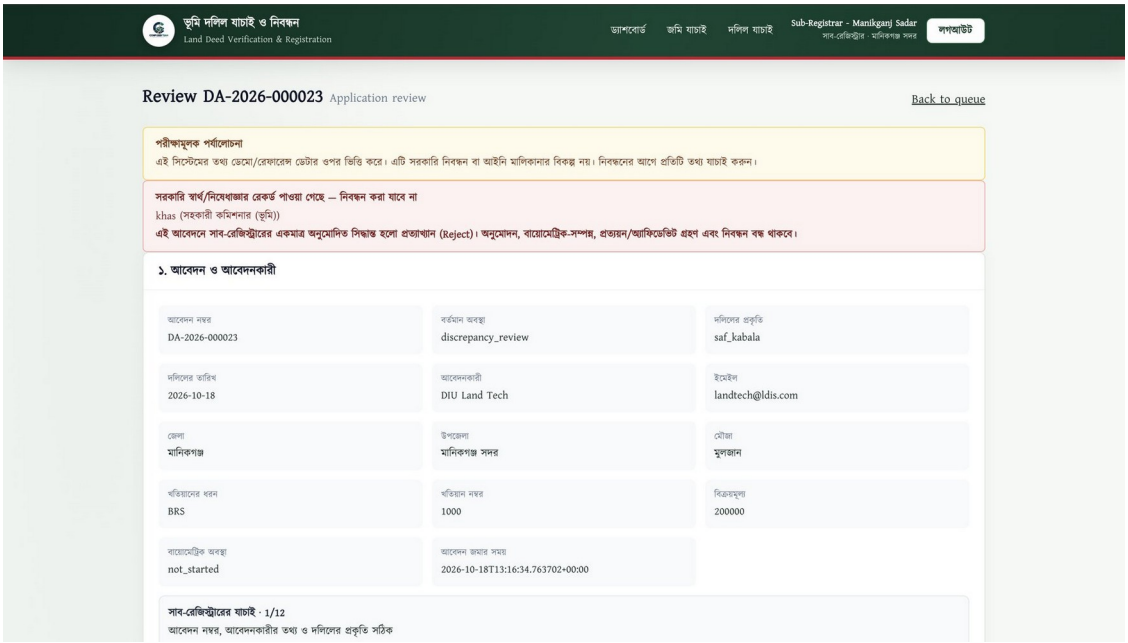


Figure 12: Government-interest hard stop message

7.10 Administrator Panel

- **User approval panel:** approve or manage newly registered users.
- **Statistics:** summary counts of applications and registrations (admin_summary).
- **All applications:** queue and detail view across jurisdictions (/admin/applications/[id]).
- **Registrar visibility:** list of Sub-Registrars with their jurisdictions.
- **Audit view:** recent audit log entries.

- **No client-side role editor:** privileged roles are provisioned only by authorised database administration, which reduces the risk of privilege abuse.

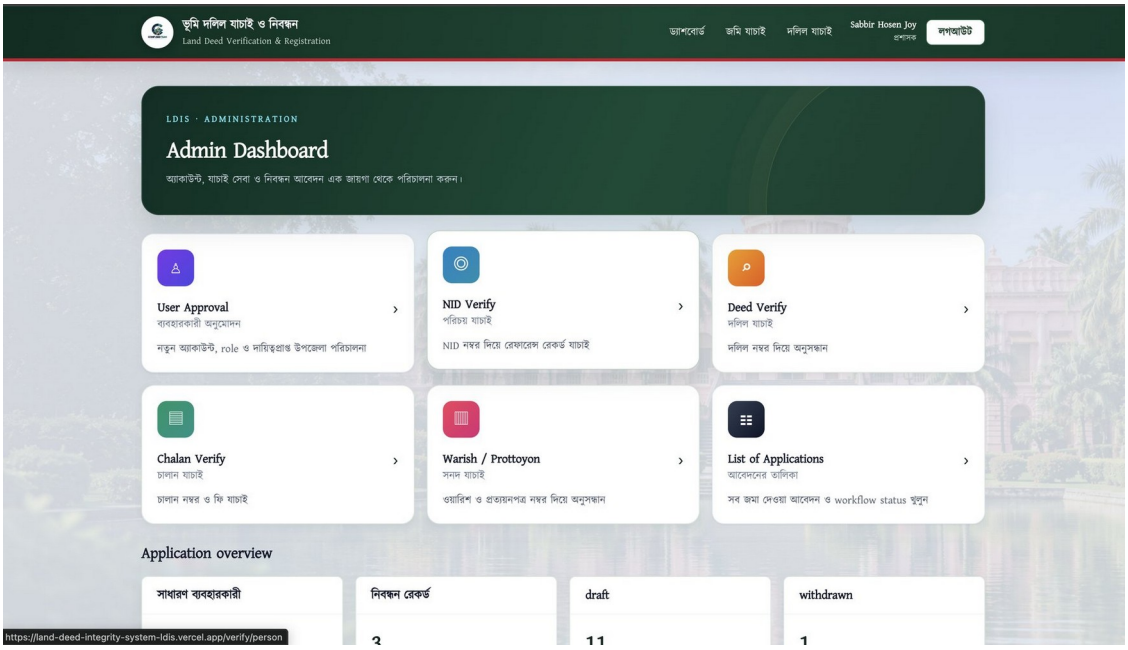


Figure 13: Administrator dashboard

7.11 Data Entry Operator Module

The data entry module (/data-entry) is restricted to the Data Entry Operator role and allows controlled, validated manual entry of reference information without direct database access.

7.12 Registered Deeds Archive and Printable Deed

After finalization, a registration record is stored in the registration register with a unique reference and a snapshot of the application. The system provides a printable page (/registered-deeds/[id]/print) with a print button. All such records are clearly labelled as DEMO records.

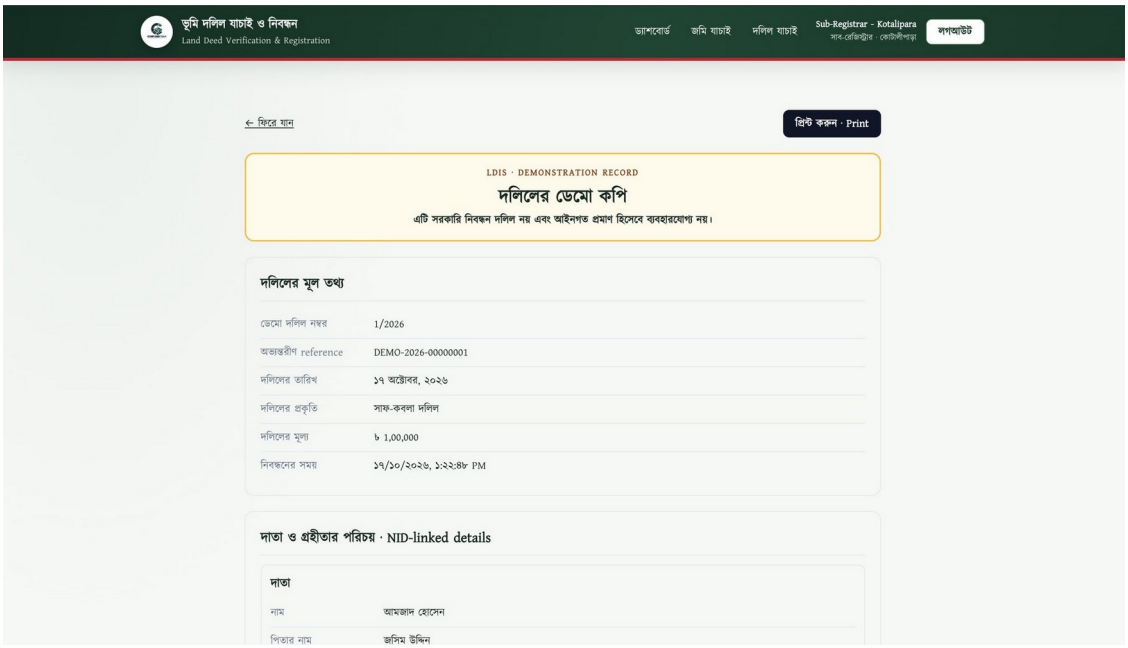


Figure 14: Printable registered deed

8. Anti-Corruption and Security Controls

This chapter explains the controls that make the system resistant to corruption. They are the main contribution of the project.

8.1 Government Land Hard Stop

The database function that detects a government interest checks two things: (a) whether any selected plot of the application matches a record in the khas table by location, khatian number and dag number, and (b) whether the application carries a computed GOVT_INTEREST_MATCH flag. If either is true, then:

- the decision function accepts only **Reject** and refuses Approve and all other decisions with the message "Government-interest record found: registration is prohibited. Only Reject is allowed.";
- the rejection must include a reason of at least five characters;
- the final registration function applies the same check before creating a registration record;
- the rule applies to both the Sub-Registrar role and the Administrator role.

Scenario. Suppose a Sub-Registrar is offered an inducement to register a plot that is recorded as government khas land. Even if the officer completes the checklist and presses Approve, the database returns an error and no registration can be created. The only available action is Reject, which is permanently logged with the officer's identity and reason. In this way the incentive for corruption is removed, because the outcome cannot be changed by the official.

8.2 Mandatory Evidence Before Approval

Table 17: Conditions that must hold before approval

Condition	Enforcement
Status is Under review	Approval from any other status is refused.
No unresolved blockers	Blockers are recomputed on the server; approval fails if any remain.
Biometric status is completed	Approval and finalization are refused until the outcome is recorded.
All 12 checklist items are Correct	Checked in the decision function and again by a trigger on the registration register.
Attestation verified (name match 50% to 90%)	Requires an uploaded attestation document and verification by the Sub-Registrar.
Affidavit received (name match below 50%)	Requires the Sub-Registrar to record receipt of the original affidavit after biometric completion.
Challans verified	Each challan must match the reference record on number, amount, type and verified status, and must not be used elsewhere.

8.3 Jurisdiction Lock

A Sub-Registrar's profile holds a district and upazila. Every staff function selects applications by matching these values with the application's office district and upazila, and raises an error if there is no match. An officer therefore cannot read or decide applications that belong to another office. A partial unique index also allows only one Sub-Registrar per upazila.

8.4 Role Integrity and Row Level Security

- Roles are read from the profiles table, not from forms or metadata.
- Signup always creates the ordinary user role.
- New users need approval before accessing the system.

- RLS is enabled on reference and sensitive tables, and no browser role is granted direct access to them.
- Staff-only evidence, such as attestation and affidavit records, is kept apart from the applicant's own view of the application.

8.5 Atomic Finalization

The finalization function runs as a single transaction. It locks the application row, validates the status, biometric result, blockers, challans and approval decision, creates a unique registration reference, inserts the registration register row, changes the status, writes the status history and the audit log, and sends a notification. If any step fails, the whole transaction is rolled back. A unique constraint prevents a second registration for the same application.

8.6 Append-Only Audit Trail

The audit log and the registration register are protected by triggers that reject any update or delete with the message "records are append-only". Every registrar decision, checklist save, verification action and finalization writes an entry containing the actor, role, action, application and details.

8.7 Other Security Measures

- Secrets are never stored in the repository; the browser uses only the public Supabase URL and the anon key.
- Safe-redirect handling prevents redirection to untrusted addresses after login.
- Server-side validation of all input and rejection of client-supplied derived values.
- Lookup attempts are logged for monitoring.
- Private storage for uploaded documents.

8.8 Residual Risks and Trust Boundaries

No system can remove every risk, and the report should be honest about this. First, the government-interest rule depends on the completeness of the khas data: when no record exists the system shows "No record found. This is NOT clearance", but it cannot know about government land that is missing from its data. Second, a person with privileged access to the database itself (the database owner) is outside the application's controls; this must be addressed by strict access management, monitoring and separation of duties. Third, the name-match threshold and some ownership rules are provisional and need legal confirmation. Fourth, the audit log protects against editing through normal operations but should be complemented by external backups and monitoring in a production deployment.

9. Testing and Validation

9.1 Automated Unit Tests

The project contains automated unit tests, run with the Node.js test runner (npm test). At the time of the final test report, 12 tests passed with 0 failures. They cover:

- application input validation and rejection of client-supplied derived values;
- application status and wizard navigation logic;
- Bangla and English digit normalization and name similarity;
- identity match outcomes and remaining-land arithmetic;
- behaviour of unknown government-interest results, challan indicators, and deed and history indicators.

The repository also contains SQL security-check scripts. The application is deployed on Vercel at the address on the title page. A full role-matrix test on a staging database (two users, two Sub-Registrars of different jurisdictions and one Administrator, including forged-role and unauthenticated calls) is recommended before any real use.

10. Deployment

- **Source control:** the project is stored in a GitHub repository.
- **Hosting:** Vercel builds and deploys the Next.js application; the vercel.json file selects the Singapore region (sin1).
- **Database:** a Supabase project holds the ref and public schemas. Migrations are applied in order, only if missing, and reset or seed scripts must never be run on the reference data.
- **Environment variables:** NEXT_PUBLIC_SUPABASE_URL and NEXT_PUBLIC_SUPABASE_ANON_KEY are configured in Vercel. No service-role key is placed in the front end or the repository.
- **Authentication settings:** the Supabase Site URL and redirect list must match the local, preview and production addresses.
- **Local run:** npm ci, copy .env.example to .env.local, set the variables, then npm run dev.

11. Limitations

LDIS is a prototype and must be described honestly. The principal limitations are:

- It uses fictional reference data and is not an official government system. It does not produce a legally valid registration.
- There is no integration with the official NID database or the national land record system.
- There is no real payment gateway; challans are verified against demo reference records.
- There is no physical biometric capture; the biometric result is recorded by the officer.
- Fees, minimum amounts, taxes and appointment calendars are demonstration values that must be confirmed with the authority.
- The 90% name-match threshold, the share-calculation assumptions and the handling of partitions and cancelled deeds are provisional.
- A missing government-interest record is not proof that the land is free of government interest.
- Complete role-based security tests against a staging database are still recommended.

12. Future Scope

Table 19: Planned improvements

Future feature	Benefit
Integration with the national NID service and official land record system	Real identity and ownership data in place of reference data.
Online payment gateway (for example bank and mobile financial services)	Direct and instantly verified payment of fees instead of manual challan matching.
Real biometric and fingerprint device integration	Genuine identity capture at the office.
OCR and AI-based document reading	Automatic extraction of data from uploaded NID, porcha and deeds, and detection of tampered documents.
Blockchain or hash-anchored deed records	Additional tamper evidence for the registration register.
QR code on deeds for public verification	Anyone can confirm that a deed is genuine.
SMS and email notifications	Immediate alerts to applicants and officers.

Future feature	Benefit
Full language toggle (Bangla and English)	Better accessibility for all users.
Complete modelling of inheritance, partition and joint ownership	More accurate entitlement and remaining-land calculation.
Anomaly detection and analytics dashboards	Identify suspicious patterns, for example an officer with unusual approval rates.
Rate limiting, monitoring and alerting	Protection against abuse and faster incident response.
Mobile application	Easier access in rural areas.
Two-person approval for high-value or flagged cases	A further separation of duties.

13. Conclusion

This project demonstrates that an important part of the integrity of land registration can be built into the system itself. By combining automated verification, explainable risk indicators, a guided application process and a database-enforced review workflow, LDIS reduces the dependence of the registration process on individual discretion. Its most important feature is that unlawful registration, such as the registration of government-owned land, is technically refused by the database even if an official is willing to approve it, while every action is recorded in a permanent audit trail.

The system is a prototype and is not ready for official use. It nonetheless provides a working model, a clear architecture and a rule set that can be extended with official data sources, payment and biometric services after legal and technical approval. The authors believe that this approach can contribute to more transparent and trustworthy land administration in Bangladesh.

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- Tailwind CSS Documentation. https://tailwindcss.com/docs
- Vercel Documentation. https://vercel.com/docs
- LDIS project documentation: PROJECT_SPEC.md, ARCHITECTURE.md, DATABASE_MAP.md, API_CONTRACTS.md, FINAL_TEST_REPORT.md (project repository).